Ex.No : 05 IMPLEMENTATION OF DATA EXTRACTION FROM AN ANDROID DEVICE USING ADB

Date :

AIM:

To extract a file from an Android emulator using Android Debug Bridge (ADB) and verify the integrity of the extracted file using the SHA-256 hashing algorithm.

SOFTWARE?EQUIPMENTS REQUIRED:

- Windows 10/11 or Linux (Ubuntu/Kali)
- Android Studio software.
- Android Debug Bridge (ADB) platform tools.

THEORY:

Android Debug Bridge (ADB) is a command-line tool that enables communication between a computer and an Android device or emulator. It consists of three components: the ADB client, ADB server, and ADB daemon (adbd). In digital forensics, ADB is used to retrieve device information, execute permitted shell commands, and transfer files for examination. The adb devices command identifies connected devices, adb push transfers files from the computer to the Android device, and adb pull extracts files from the device to the computer. In this experiment, an Android emulator is used to demonstrate file transfer and extraction. The SHA-256 cryptographic hash algorithm is then applied to the original and extracted files to compare their hash values and verify file integrity. Matching hash values indicate that the file contents are identical with respect to the SHA-256 comparison. Proper documentation, authorization, and preservation of original evidence are essential for maintaining forensic reliability.

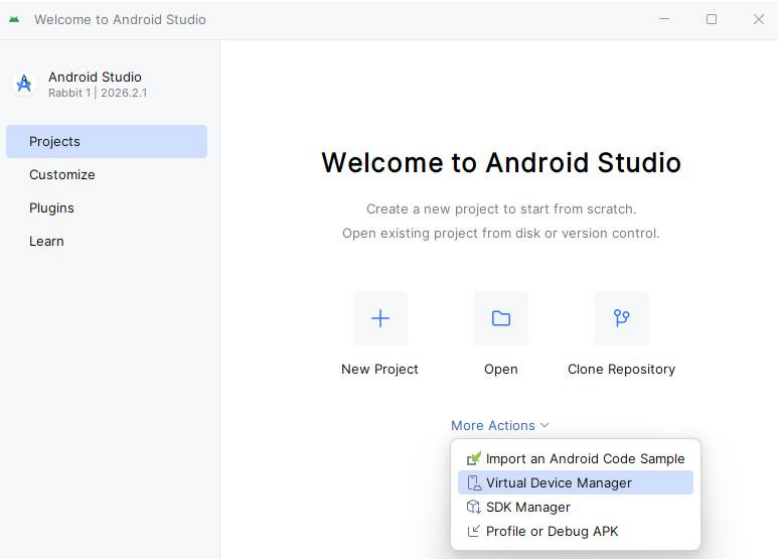
ALGORITHM:

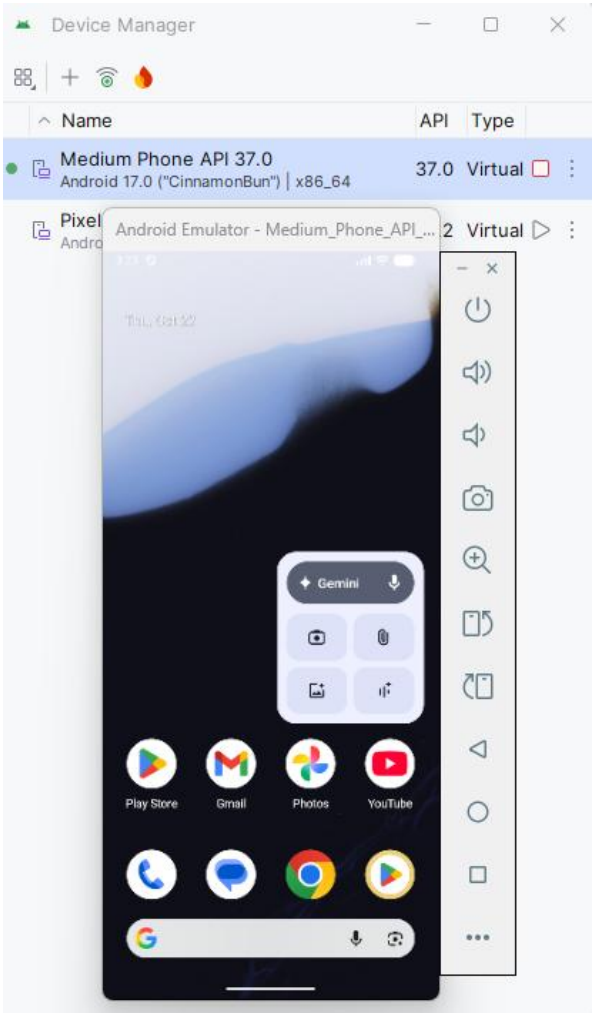
1. Download and install ExifTool.
2. Start the Android emulator using Android Studio.
3. Verify the ADB installation and emulator connection.
4. Retrieve the Android device information using ADB commands.
5. Create a sample text file on the computer.
6. Transfer the sample file to the Android emulator using the `adb push` command.
7. Verify the presence of the file on the emulator.
8. Extract the file from the emulator to the computer using the `adb pull` command.
9. Calculate the SHA-256 hashes of the original and extracted files.
10. Compare both hash values to verify file integrity.
11. End

PROCEDURE

Step 1: Start the Android Emulator

1. Open Android Studio.
2. Navigate to Tools → Device Manager.
3. Create a virtual Android device if one is not available.
4. Start the emulator.





Step 2: Verify ADB Installation

Open Command Prompt and execute:

adb version

This command displays the installed ADB version.

Change directory if needed: cd C:\Users\dhamu\AppData\Local\Android\Sdk\platform-tools

Step 3: Check Device Connection

Execute: adb devices

Verify that the emulator appears in the device list with the status device.

Step 4: Collect Device Information

Execute the following commands:

adb shell getprop ro.build.version.release

adb shell getprop ro.product.model

adb shell getprop ro.product.manufacturer

These commands retrieve the Android version, device model, and manufacturer.

```
Microsoft Windows [Version 10.0.26200.9711]
(c) Microsoft Corporation. All rights reserved.

C:\Users\dhamu>cd C:\Users\dhamu\AppData\Local\Android\Sdk\platform-tools

C:\Users\dhamu\AppData\Local\Android\Sdk\platform-tools>adb version
Android Debug Bridge version 1.0.41
Version 37.0.1-15733141
Installed as C:\Users\dhamu\AppData\Local\Android\Sdk\platform-tools\adb.exe
Running on Windows 10.0.26200

C:\Users\dhamu\AppData\Local\Android\Sdk\platform-tools>adb devices
List of devices attached
emulator-5554   device


C:\Users\dhamu\AppData\Local\Android\Sdk\platform-tools>adb shell getprop ro.build.version.release
17

C:\Users\dhamu\AppData\Local\Android\Sdk\platform-tools>adb shell getprop ro.product.model
sdk_gphone64_x86_64

C:\Users\dhamu\AppData\Local\Android\Sdk\platform-tools>adb shell getprop ro.product.manufacturer
Google

C:\Users\dhamu\AppData\Local\Android\Sdk\platform-tools>
```

Step 5: Create a Sample File

Execute:

mkdir "%USERPROFILE%\Documents\MobileForensicsLab"

echo Digital Forensics Lab Sample >

"%USERPROFILE%\Documents\MobileForensicsLab\sample.txt"

This creates a sample text file on the computer.

Step 6: Transfer the File to the Emulator

Execute:

adb push "%USERPROFILE%\Documents\MobileForensicsLab\sample.txt" /sdcard/Download/sample.txt

The file is copied from the computer to the emulator.

Step 7: Verify the File on the Emulator

Execute:

adb shell ls -l /sdcard/Download/sample.txt

adb shell cat /sdcard/Download/sample.txt

These commands verify the file's presence and display its contents.

Step 8: Extract the File from the Emulator

Execute:

adb pull /sdcard/Download/sample.txt "%USERPROFILE%\Documents\MobileForensicsLab\extracted_sample.txt"

The file is copied back from the emulator to the computer.

Step 9: Verify File Integrity

Calculate the SHA-256 hash of the original file:

certutil -hashfile "%USERPROFILE%\Documents\MobileForensicsLab\sample.txt" SHA256

Calculate the SHA-256 hash of the extracted file:

certutil -hashfile "%USERPROFILE%\Documents\MobileForensicsLab\extracted_sample.txt" SHA256

Compare the two hash values. If both values match, the original and extracted files have identical contents with respect to the SHA-256 comparison.

OUTPUT:

SHA-256 hash of the original file:

SHA-256 hash of the extracted file: